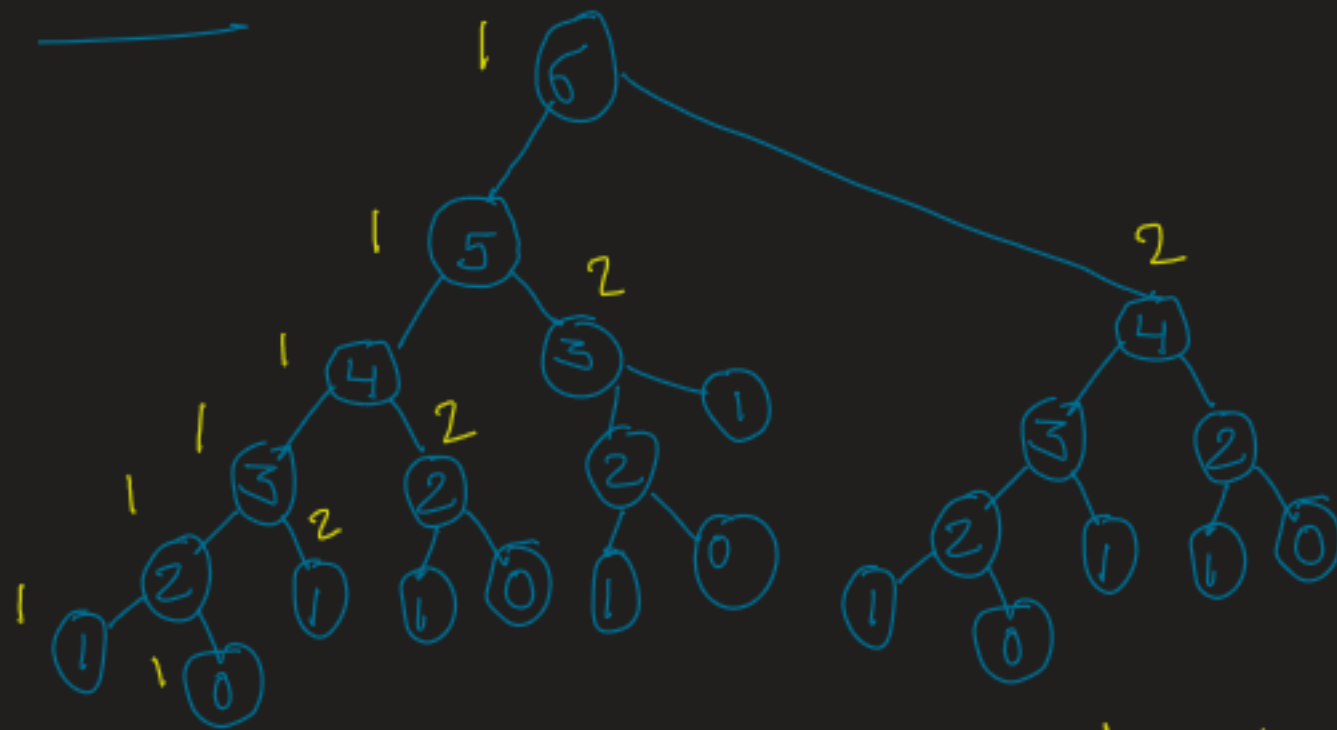


* Detailed View on Time and Space Complexity:-

~ Memorization (Top Down)

Time $\rightarrow O(2^*N)$

$n=6$



$\rightarrow$ Any value could be visited only 2 Times in the worst case.

Thus Time is $O(2^*N)$

* Bottom-up :-

Time $\rightarrow O(3^{\text{given-}n})$

$\rightarrow O(3^n)$, let given- $n = n$

$n = 7$

dp =

visitations $\rightarrow$

0	1	2	3	4	5	6	7
0	1	1	2	3	5	8	13
1	1						
2	2	1					
2	3	2	1				
2	3	3	2	1			
2	3	3	3	2	1		
2	3	3	3	3	2	1	
2	3	3	3	3	3	2	1

→ Any value could be visited only
3 Times in the worst case.

Thus Time is $O(3^*N)$

→ Memoization is taking less time in this it
does not mean Bottom up is inefficient.
This is simple Problem with less calculations.
In future you will see how efficient
Bottom-up is compared to Memoization.